NEXTEST™ SYSTEMS CORPORATION

HostmonView (Magnum V and EPIC)

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Revision History

Date	Type	Details
4/12/05	---	Initial Document
7/15/05	Update	Substantial updates based on implementation changes.
11/27/11	Update Add	Substantial updates based on implementation changes to the following: • Basic View Display - GUI changes. • DC Power On/Off Controls (Pre-Firmware 7.4.1) - new. • DC Power ON/OFF Procedures - new, required.
1/9/15	Update	First Substantial Magnum V Update
7/5/16	Update	Added Power Lock Mode description
7/21/17	Update	Added new section and updated screenshots for new DUT Board Power Indicators
5/4/20	Update	Adding 3-section HostMonView GUI introduced with APB/SPC Firmware Version 7.4.1.

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Chapter 1 HostmonView

This document contains the following:

- Overview
- Communications Architecture
- Starting HostmonView
- HostmonView Color Codes
- Basic View Display
- DC Power On/Off Controls (Pre-Firmware 7.4.1)
- DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up)
- Detailed View Display
- Prober View Display
- ZIF View Display

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1.1 Overview

Note: The DC Power ON/OFF Procedures must be used to turn DC power OFF to the site controllers and DUTBOARDs *before* using the hardware switches to turn OFF AC power. This is required for system reliability.

HostmonView is a graphical interface allowing the user to control system/card cage DC power, view the overall system status and, optionally, view the following information from the APB Computer in each card cage (see Communications Architecture):

- Which APB Computers are on-line and communicating with the hostmon process.
- Which Site Controllers are on-line and communicating with the hostmon process.
- If TIU POWER is ON or OFF for a particular APB.
- If DUTBOARD POWER is ON or OFF for a particular APB.
- Various monitored and measured power supply voltages, board temperatures and fan speeds.

Note: A 3-section HostMonView GUI was introduced with APB/SPC Firmware Version 7.4.1. If a Tester is using APB/SPC Firmware Version 7.4.1 or greater, this 3-section GUI will be displayed. This document displays both the pre-7.4.1 and the post 7.4.1 version of the GUI.

HostmonView is normally started automatically (see Starting HostmonView) and is represented by an icon in the system tray (see HostmonView System Tray Icon). Double-clicking on the tray icon causes the HostmonView GUI to display. It has two display options:

- Basic View Display - the initial display, which reflects the overall system status, presents the system-level DC Power On/Off Controls (Pre-Firmware 7.4.1) or DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up) depending on the Firmware version (see the note above), and displays green indicators for the APBs and site controllers that are currently responding to TCP/IP communications. This is suitable for all users, operators, technicians, etc.

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- Detailed View Display - this is targeted at maintenance activities, allowing the user to view details of a selected card cage's and Site Assembly's DC power supply voltages, board temperature, etc. In this view, HostmonView will also display the source of any faults detected by an APB, any status messages generated by and APB, and any status messages generated by a Site Assembly's Site Controller.
- Prober View Display - Used only in Probe Systems to operate the Prober interface to the Test System.
- ZIF View Display - Used to operate the ZIF connectors that connect the Site Assemblies with the Tester Interface Unit (TIU).
- Always on Top - Clicking this button keeps HostmonView on top of the other screens of the Host Computer.
  Its display is invoked by:
  Clicking the fifth button of the HostmonView Tool Bar in the Basic View Display:

- Legend (?) - Clicking this button displays the Legend.

A Magnum system may consist of 1 to 4 card cages. Each card cage contains from 1 to 10 Site Assemblies, each with a local Site Controller.

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1.2 Communications Architecture

See Overview.

The following diagram shows the key software processes and communications architecture related to HostmonView. This is provided for maintenance personnel to aid in understanding how the APB and HostmonView operate:

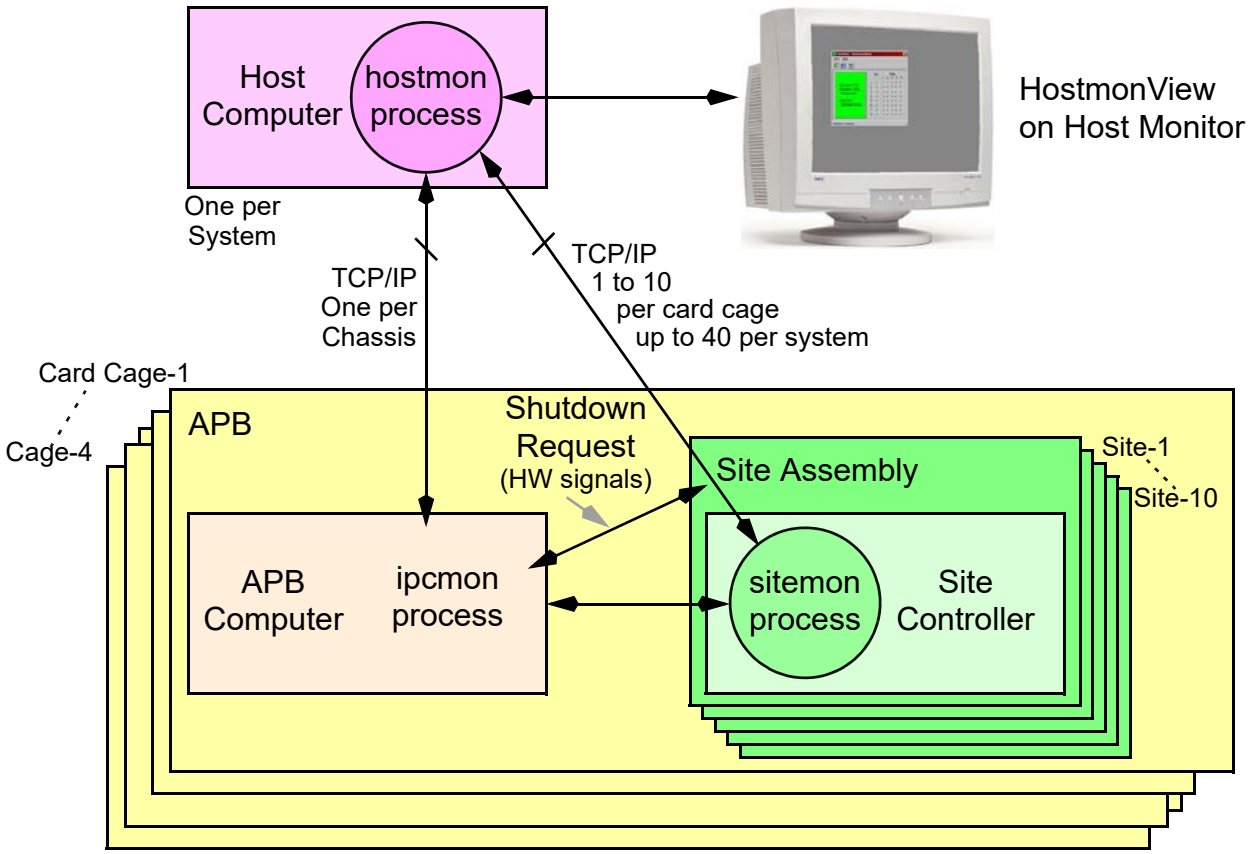

Figure-1: HostmonView Processes & Communications Architecture

Note the following:

- HostmonView directly communicates with the hostmon process executing on the Host Computer. *hostmon* communicates with the APB Computer in each card cage and the sitemon process executing on each Site Controller.

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- The hostmon process is started by *UseRel.bat*, normally executed automatically when a user logs on to the Host computer. The sitemon process starts when a given Site Controller completes its boot sequence. The hostmon process also starts HostmonView, which is represented by an icon in the system tray (see HostmonView Color Codes).
- If a given APB detects a critical temperature or voltage fault, it disables power to the Site Assembly or Site Assemblies depending on the fault that occurs. The card cage's APB Computer then sends a notification to the hostmon process that the Site Assembly or Site Assemblies were powered-down. *HostMonView* then queries the APB Computer to determine the nature of the fault. The source can be a Site Assembly over-temperature fault. The Site Assembly or Site Assemblies will generate a Shutdown Request if its local temperature or voltage sensors detect a fault or a local DC-to-DC converter detects a fault. Note that the APB does not know why a given Site Assembly requests a Shutdown Request, thus it can't inform the user with a status message.

1.3 Starting HostmonView

See Overview.

HostmonView is normally started automatically, by *hostmon*, which is itself started by *UseRel.bat*, which is normally automatically executed when the user logs-in to the host computer.

HostmonView can also be [re]started by double-clicking the *hostmon* icon in the system tray:

Figure-2: HostmonView System Tray Icon

HostmonView will initially show the Basic View Display. Note that if the HostmonView icon is not displayed in the system tray the user must execute *UseRel.bat* from an appropriate Nextest software release to start the hostmon process.

If Test System power is OFF, the System Tray icon and the Basic View Display will be grey.

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Note: *hostmon* and HostmonView both use the same tray icon. The icon is displayed as long as the hostmon process is running, regardless whether HostmonView is running. Double-clicking on the icon will start HostmonView if it is not already running.

1.4 HostmonView Color Codes

See Overview.

HostmonView uses five colors to indicate status. These apply to the tray icon and to the Status Icon shown in the Basic View Display, as shown below:

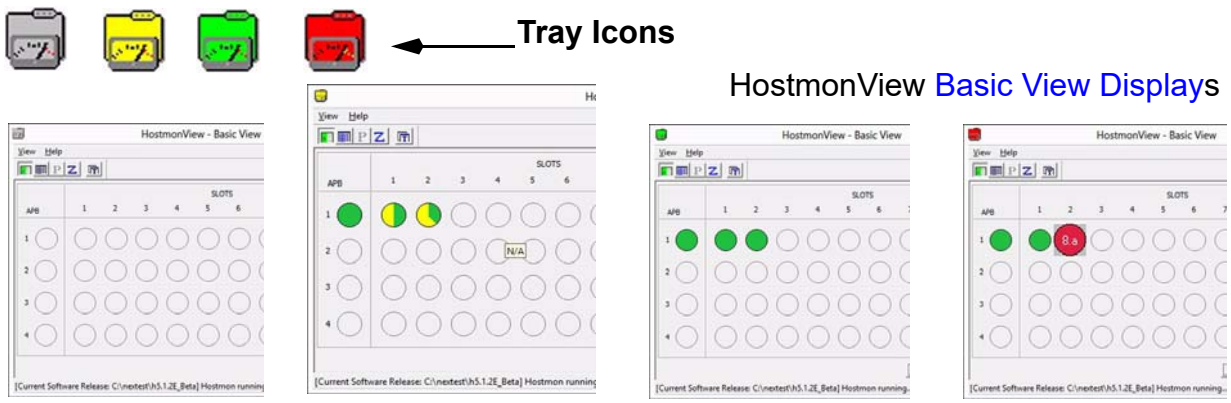

Figure-3: HostmonView Color Codes

- Gray = This only applies to the System Tray icon. It means that the APBs are not running. It usually indicates that Test System power is OFF. See Starting HostmonView.
- Yellow = This usually means that test head power is ON but that none of the Site Controller(s) have completed their boot sequence far enough to establish TCP/IP communications with the hostmon process. This is signified by the half-yellow, half-green Site Computer icons shown in the figure above. The other case where the Tray Icon is yellow is when the APBs are off.
- Green = One or more APB Computer(s) are communicating with the hostmon process AND the Site Controller(s) with green Site Computer Icons are communicating with the hostmon process. This also means that no critical faults

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are currently detected by an APB Computer which, if a fault was detected, will power-down the affected site controller, causing HostmonView to display the Site Controller's icon as Red with a lightning bolt inside it.

- Red (with "MODULE#.ERROR#" label) = The red Site Controller(s) are communicating with the hostmon process but have reported a critical fault and have powered-down. If HostmonView's Basic View Display is not visible click on the tray icon to display it to determine which Site Controller reported the fault.

Note: The significance of the color of the various TABs displayed in the Detailed View Display is somewhat different than the descriptions above. See Detailed View Display.

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1.5 Basic View Display

When the *hostmon* tray icon is clicked it will display HostmonView's basic view dialog, as shown below. This view is suitable for constant display in a production environment, allowing the production operator to immediately notice when the system has detected a fault.

Note: Figure-4: Site and DUTBOARD Power ON/OFF Switches (Pre 7.4.1 Version) displays the Basic View for the Pre APB/SPC 7.4.1 Firmware in Magnum V. Figure-5: DC Power Controls Visible (Magnum EPIC) displays the Basic View for APB/SPC 7.4.1 and Up Firmware.

1.5.1 DC Power On/Off Controls (Pre-Firmware 7.4.1)

Note: The DC Power ON/OFF Procedures must be used to turn DC power OFF to all sites and DUTBOARDs *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

Note: These controls **ONLY** affect the DC power to the sites and DUTBOARDs. They do NOT control the power to the rest of the Test System, which remains ON until a hardware switch or circuit breaker is used to turn the AC power OFF.

By default, HostmonView's Basic View Display shows the Site DC Power Switches and DUTBOARD Power Switches. These may be hidden using the DC Power Controls Hide/

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Show Button shown in Figure-4: Site and DUTBOARD Power ON/OFF Switches (Pre 7.4.1 Version) below.

Site DC Power Switches

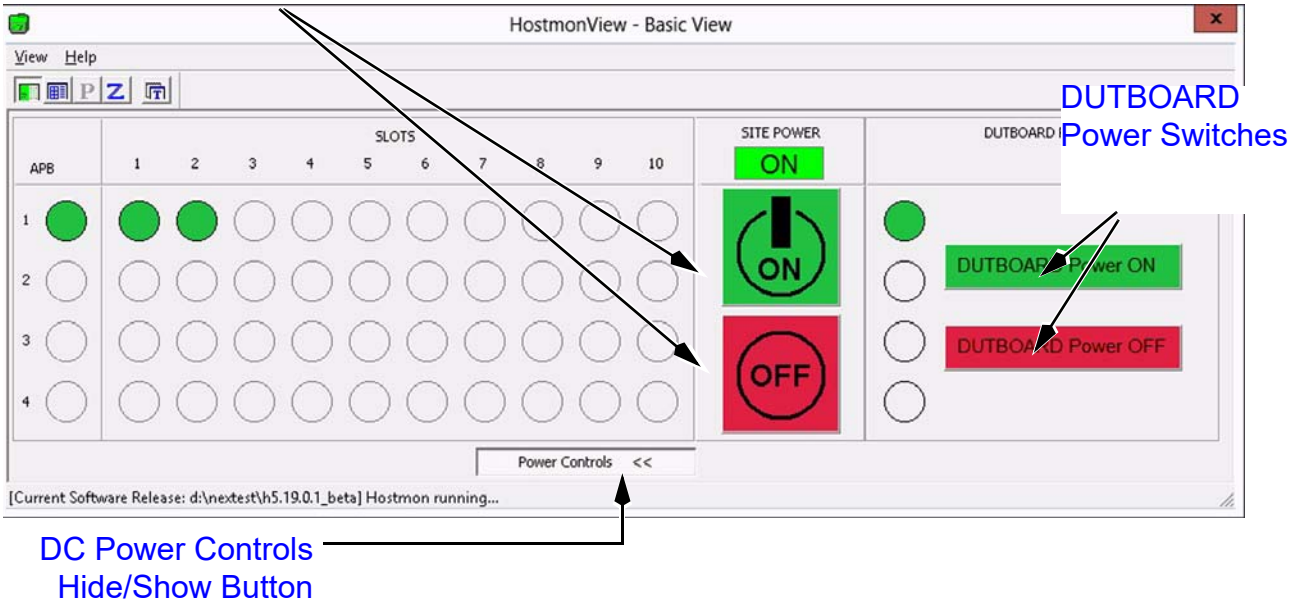

Figure-4: Site and DUTBOARD Power ON/OFF Switches (Pre 7.4.1 Version)

When displayed, the following controls and indicators are used to examine and manipulate power to individual chassis or chassis-pairs.

Note: The following components displayed in the figure above apply to the HostMonView GUI used with APB/SPC Firmware versions previous to 7.4.1:

Item	Description
Site DC Power Switches	Used to turn DC power ON or OFF all DC power supplies in the sites (i.e. all chassis at one time).
DUTBOARD Power Switches	Used to turn DC power ON or OFF to the DUTBOARDS.
DC Power Controls Hide/Show Button	Used to expose (show) or hide the Site DC Power Switches and the DUTBOARD Power Switches.

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1.5.1.1 DC Power ON/OFF Procedures

See Overview, Basic View Display, DC Power On/Off Controls (Pre-Firmware 7.4.1).

Note: Using Magnum V, the DC Power ON/OFF Procedures must be used to turn DC power OFF to all sites and DUTBOARDs *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

Note: These controls **ONLY** affect the DC power to the sites and DUTBOARDs. They do NOT control the power to the rest of the Test System, which remains ON until a hardware switch or circuit breaker is used to turn the AC power OFF.

Turning on Site DC Power causes each site controller to begin its boot sequence. The sites are ready to use when a VNC window has opened for its site controller. A VNC window must open for each site controller in the system that will be used for testing.

The following procedures are documented below and use DC Power On/Off Controls (Pre-Firmware 7.4.1):

- Site DC Power ON Procedure
- Site DC Power OFF Procedure

Site DC Power ON Procedure

See DC Power ON/OFF Procedures.

The following procedure presumes AC Power is ON in the system and that DC power is OFF in the site controllers.

To turn DC power ON to the site controllers:

1. If the HostmonView dialog is not displayed, double-click the HostmonView icon in the system tray to display the dialog:
2. Click the Site Power On Switch to turn ON the DC power to the site controllers:

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. When this is done, a dialog similar to the following will be presented:

To complete the operation click the **YES** button, or click **Cancel** to cancel the operation.

If the **YES** button is selected, the following dialog will be displayed for a period of time:

Note the following:

- Turning Site power ON causes every active site controller in the Test System to begin its boot sequence, which can take some time to complete. During this time, the Status Icon and Site Status Indicators will slowly, and at different times, change color (state) as the boot process proceeds. But...

Note: If the YELLOW part of the slot icon changes to ORANGE, this means that the Site is in Power Lock Mode.Please wait until the ORANGE part of the Site Icon turns to either back to YELLOW or to GREEN. If the Site Icon does not turn either YELLOW or to GREEN, see Power Lock Mode below for instructions. The Test System is not ready for use until all of these indicators have turned green.

Caution – Powering off site power during Power Lock Mode could severely damage the Site Assemblies and make them nonoperational.

Site DC Power OFF Procedure

See DC Power ON/OFF Procedures.

The following procedure presumes AC and DC power are currently ON in the Test System and that DC Power is ON in the site controllers.

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To turn DC power OFF to the site controllers:

1. If the HostmonView dialog is not displayed, double-click the HostmonView icon in the system tray to display the dialog:

2. Click the Site Power Off Switch to turn OFF the DC power to the site controllers:

   . When this is done, a dialog similar to the following will be presented:

   To complete the operation click the **`Power OFF`** button, or click **`Cancel`** to cancel the operation.

   If the **`Power OFF`** button is selected, the following dialog will be displayed and remain for a short period of time:

   The Site Power Indicator turns red and its label text changes to OFF.

Note: The AC Breakers should only be turned OFF <u>after</u> the site controller power has been turned OFF.

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1.5.1.2 DUTBOARD Power ON/OFF Procedures

See Overview, Basic View Display, DC Power On/Off Controls (Pre-Firmware 7.4.1).

Note: These controls **ONLY** affect the DC power to the DUTBOARDs, they do NOT control the AC power or Test System power.

Note: DUTBoard power is not controlled with HostMonView in Test Systems that use a Prober.

The following procedures are documented below and use DC Power On/Off Controls (Pre-Firmware 7.4.1):

- DUTBOARD DC Power ON Procedure
- DUTBOARD DC Power OFF Procedure

DUTBOARD DC Power ON Procedure

See DUTBOARD Power ON/OFF Procedures.

The following procedure presumes that DUTBOARD power is OFF.

To turn DUTBOARD power ON:

1. If the HostmonView dialog is not displayed, double-click the HostmonView icon in the system tray to display the dialog:
2. Click the DUTBOARD Power On Switch to turn ON the DUTBOARD power.

   The DUTBOARD Power Indicators turns green and its label text changes to ON.

DUTBOARD DC Power OFF Procedure

See DUTBOARD Power ON/OFF Procedures.

The following procedure presumes DUTBOARD power is currently ON in the system.

To turn DUTBOARD power OFF:

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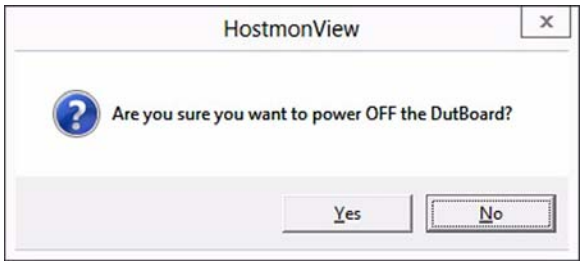
1. If the HostmonView dialog is not displayed, double-click the HostmonView icon in the system tray to display the dialog:

2. Click the DUTBOARD Power Off Switch to turn OFF the DC power to the DUTBOARDs. . When this is done, a dialog similar to the following will be presented:



To complete the operation click the **`Yes`** button, or click **`Cancel`** to cancel the operation. The DUTBOARD Power Indicators turns red and its label text changes to OFF.

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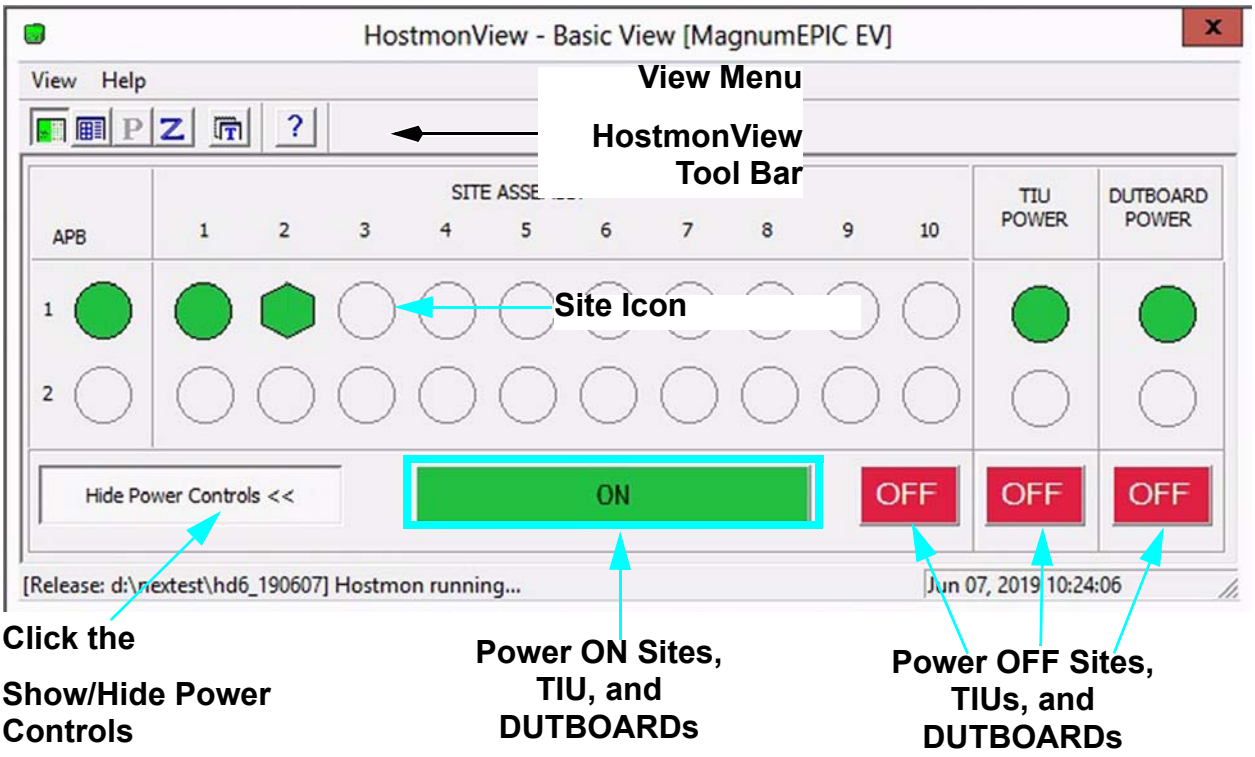
1.5.2 DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up)

When the Test System powers but the Site Power is still OFF, the following Basic View Dialog is displayed:

Figure-5: DC Power Controls Visible (Magnum EPIC)



The dialog below is displayed when the Show/Hide Power Controls button is clicked to hide the Power Controls:

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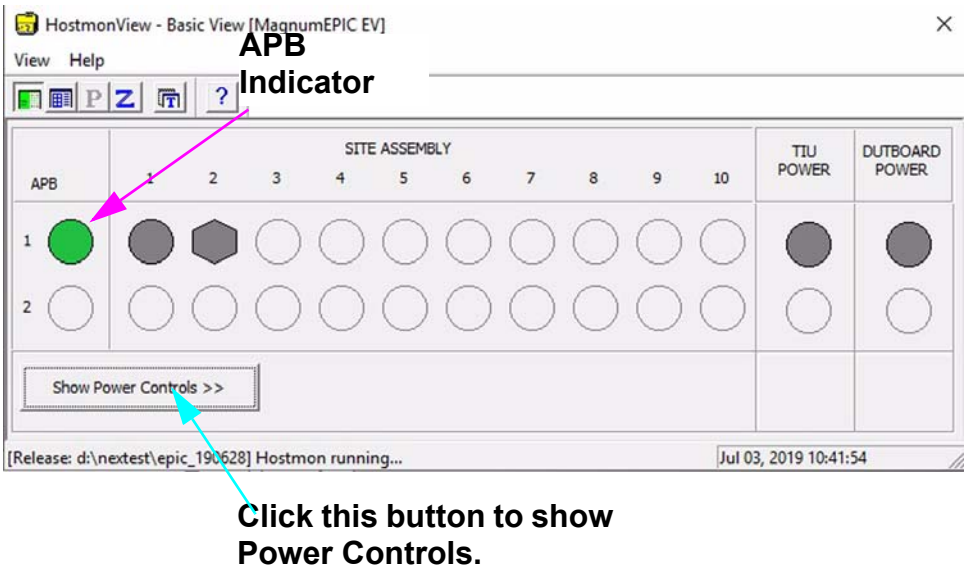
Figure-6: DC Power Controls Hidden (Magnum EPIC)



Note: The DC Power ON/OFF Procedures must be used to turn DC power OFF to all sites, TIUs, and DUTBOARDs *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

Note: These controls **ONLY** affect the DC power to the sites, TIUs, and DUTBOARDs. They do NOT control the power to the rest of the Test System, which remains ON until a hardware switch or circuit breaker is used to turn the AC power OFF.

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By default, HostmonView's **Basic View Display** shows the DC Power Controls. These may be hidden using the DC Power Controls Hide/Show Button shown in Figure-7: Site, TIU, and DUTBOARD Power ON/OFF Switches (7.4.1 & Up Version) below.

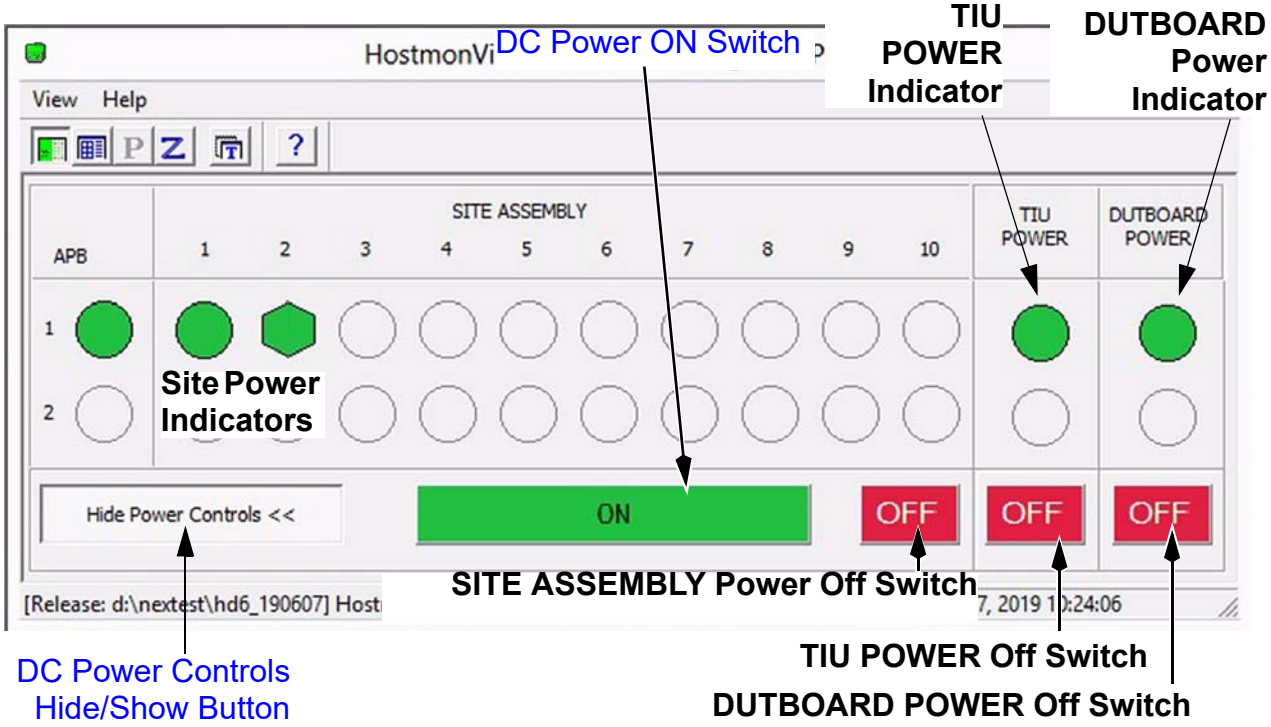

Figure-7: Site, TIU, and DUTBOARD Power ON/OFF Switches (7.4.1 & Up Version)

When displayed, the following controls and indicators are used to examine and manipulate power to individual chassis or chassis-pairs.

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Note: The following components displayed in the figure above apply to the HostMonView GUI introduced in APB/SPC Firmware Version 7.4.1:

Item	Description
DC Power ON Switch	Opens a dialog box that is used to turn on Site, TIU, and DUTBOARD Power.
DC Power OFF Switches	Used to turn **SITE ASSEMBLY**, **TIU**, or **DUTBOARD** DC power OFF. See SITE ASSEMBLY Power Off Switch, TIU POWER Off Switch, and DUTBOARD POWER Off Switch for illustrations.
DC Power Controls Hide/Show Button	Used to expose (show) or hide the DC Power ON Switch and the DC Power OFF Switches shown in the figure above.

1.5.2.1 DC Power ON/OFF Instructions (Firmware 7.4.1 & Up)

Note: Using Magnum V, the DC Power ON/OFF Procedures must be used to turn DC power OFF to all DUTBOARDs, TIUs, and Sites *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

Note: These controls **ONLY** affect the DC power to the sites and DUTBOARDs. They do NOT control the power to the rest of the Test System, which remains ON until a hardware switch or circuit breaker is used to turn the AC power OFF.

Turning on Site DC Power causes each site controller to begin its boot sequence. The sites are ready to use when a VNC window has opened for its site controller. A VNC window must open for each site controller in the system that will be used for testing.

The following procedures are documented below and use DC Power On/Off Controls (Pre-Firmware 7.4.1):

- DC Power ON Instructions
- DUTBOARD POWER OFF Instructions
- TIU POWER OFF Instructions

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- SITE (ASSEMBLY) Power OFF Instructions

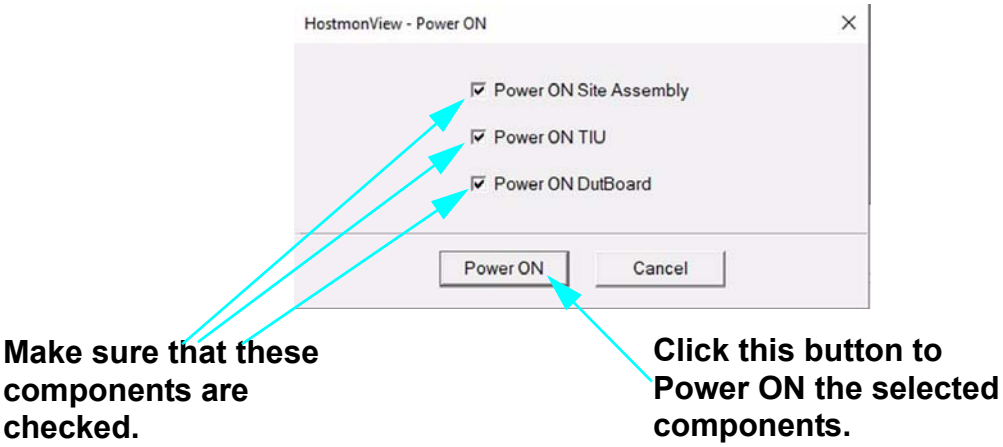
DC Power ON Instructions

The following procedure presumes AC Power is ON in the system and that DC power is OFF in the site controllers(, TIU, and DUTBOARD if applicable).

To turn on Site, TIU, and DUTBOARD Power:

1. If the HostmonView dialog is not displayed, double-click the HostmonView icon in the system tray to display the dialog:
2. Click ON (DC Power ON Switch) to display the HostmonView Power ON Dialog (next page).
3. Make sure the **Power ON Site Assembly**, **Power ON TIU**, and **Power ON DutBoard** check boxes are selected on the screen that appears, then click **Power ON**.

Figure-8: HostmonView Power ON Dialog



This will start the power up sequence for all three components or if one of the components already started, it will start the remaining ones.

Note the following:

- The color of the Status Icon indicates the overall status of the APBs and the site controllers. See TIU/DUTBOARD Power Indicator Legend.
- The APB Status Indicators will be lit for any APB Computer which has established TCP/IP communications with the hostmon process. The indicator will be Green when the APB is working properly.

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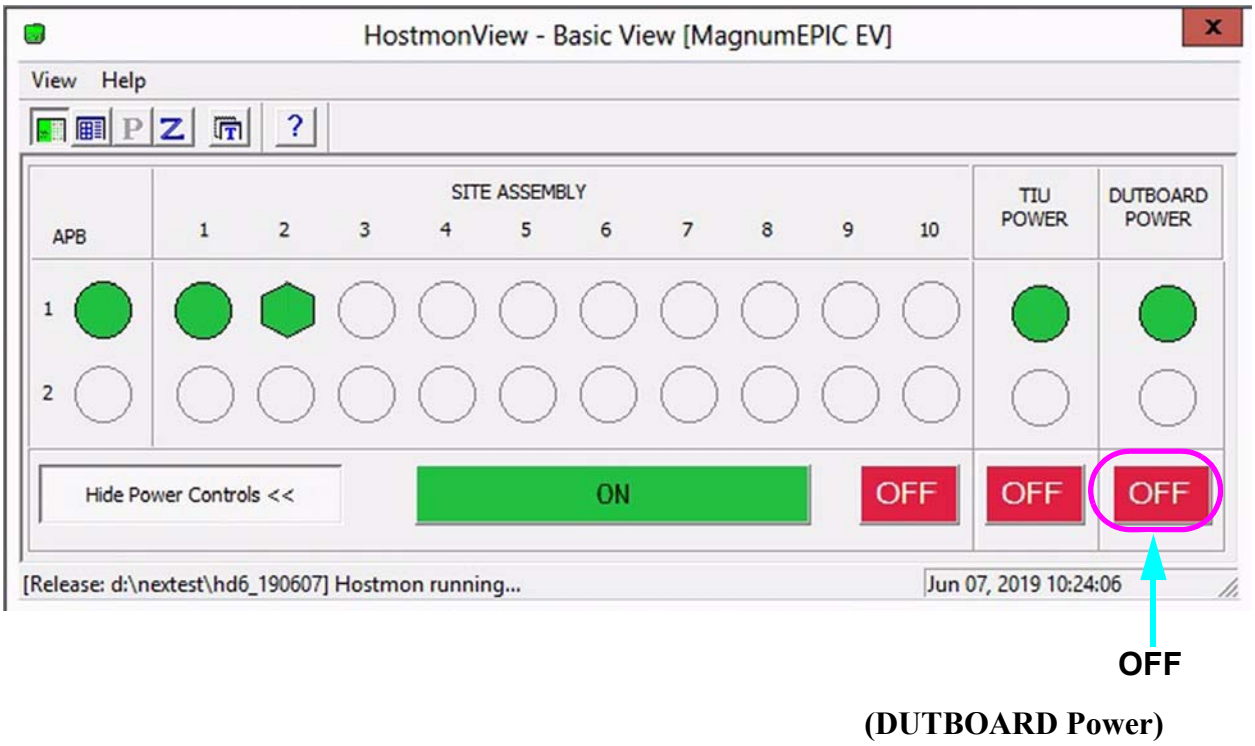
DUTBOARD POWER OFF Instructions

Note: DC power be turned OFF first to all DUTBOARDs (using the instructions in this section), then to all TIUs, and Sites *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

To turn DUTBOARD POWER OFF:

1. Click the OFF button in the DUTBOARD POWER section of Figure-9: DUTBOARD Power Off in HostmonView (Magnum EPIC Example) below.

Figure-9: DUTBOARD Power Off in HostmonView (Magnum EPIC Example)



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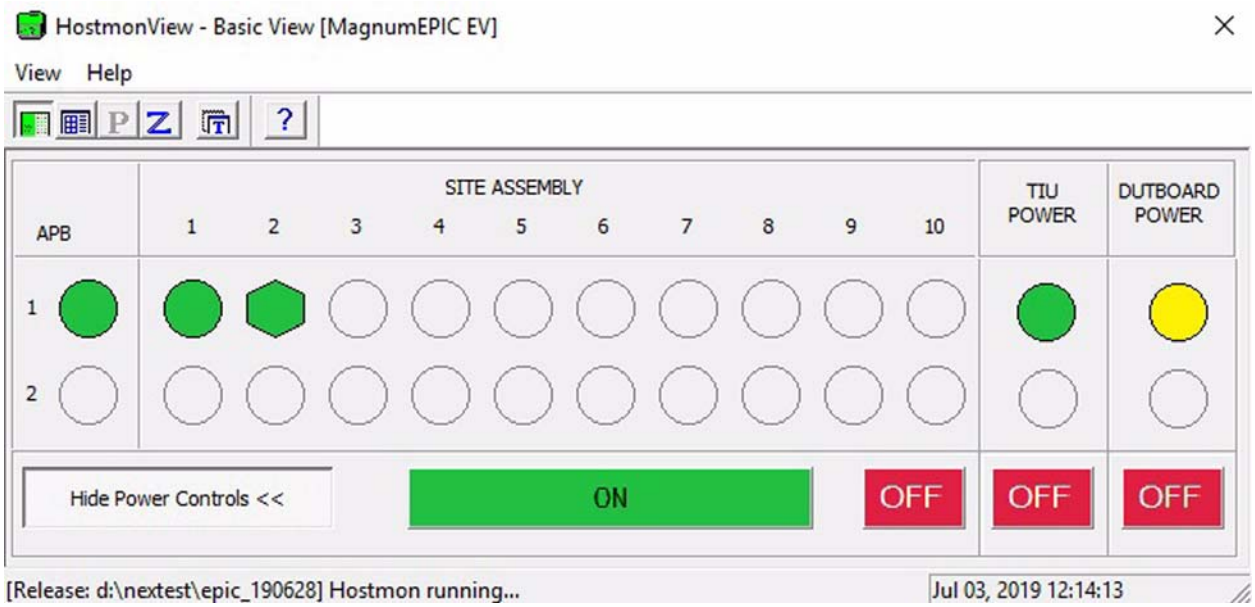
The DUTBOARD POWER indicator(s) will display a yellow background in the DUTBOARD POWER section. This means that DUTBOARD power is off. See Figure-10: DUTBOARD Power OFF (Magnum EPIC Example) below. See Figure-18: TIU/ DUTBOARD Power Indicator Legend, page 33 for other possible DUTBoard Power status.

Figure-10: DUTBOARD Power OFF (Magnum EPIC Example)



TIU POWER OFF Instructions

Note: If the system being turned OFF is a non-VU Magnum V system, there is no need to perform these instructions. In fact, the button used to perform them will not even appear in HostMonView.

DC power must be turned OFF first to all DUTBOARDs, then to all TIUs (using the instructions in this section), and Sites *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

To turn TIU POWER OFF:

1. Click the OFF button in the TIU POWER section of Figure-11: TIU Power Off in HostmonView (Magnum EPIC Example) below.

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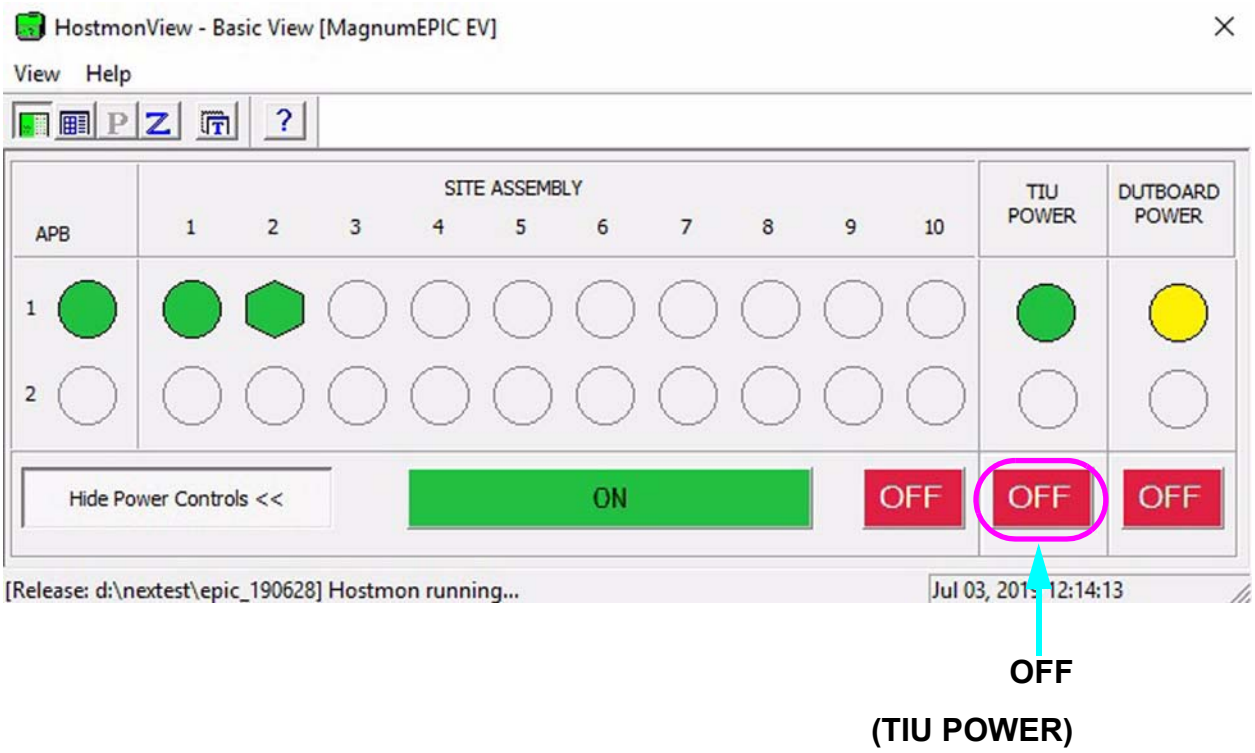
Note: If DUTBOARD POWER is still ON, turning TIU POWER OFF will also turn off DUTBOARD POWER.

Figure-11: TIU Power Off in HostmonView (Magnum EPIC Example)



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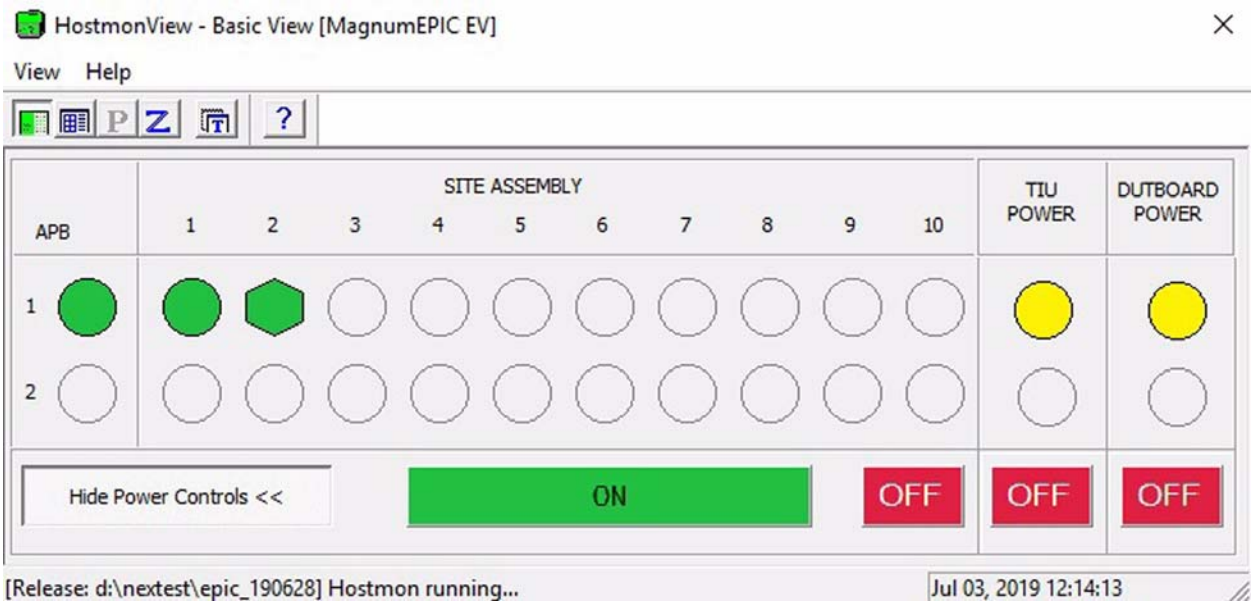
The TIU POWER indicator will display a yellow background in the TIU POWER section. This means that TIU power is off. See Figure-12: TIU Power OFF (Magnum EPIC Example) below. See Figure-18: TIU/DUTBOARD Power Indicator Legend, page 33 for other possible DUTBoard Power status.

Figure-12: TIU Power OFF (Magnum EPIC Example)



SITE (ASSEMBLY) Power OFF Instructions

Note: DC power must be turned OFF first to all DUTBOARDs, then to all TIUs, and Sites (using the instructions in this section) *before* using any hardware switches or circuit breakers to turn OFF power to the Test System. This is required for system reliability.

To turn SITE (ASSEMBLY) POWER OFF:

1. Click the OFF button in the SITE ASSEMBLY section of Figure-13: SITE Power Off in HostmonView (Magnum EPIC Example) below.

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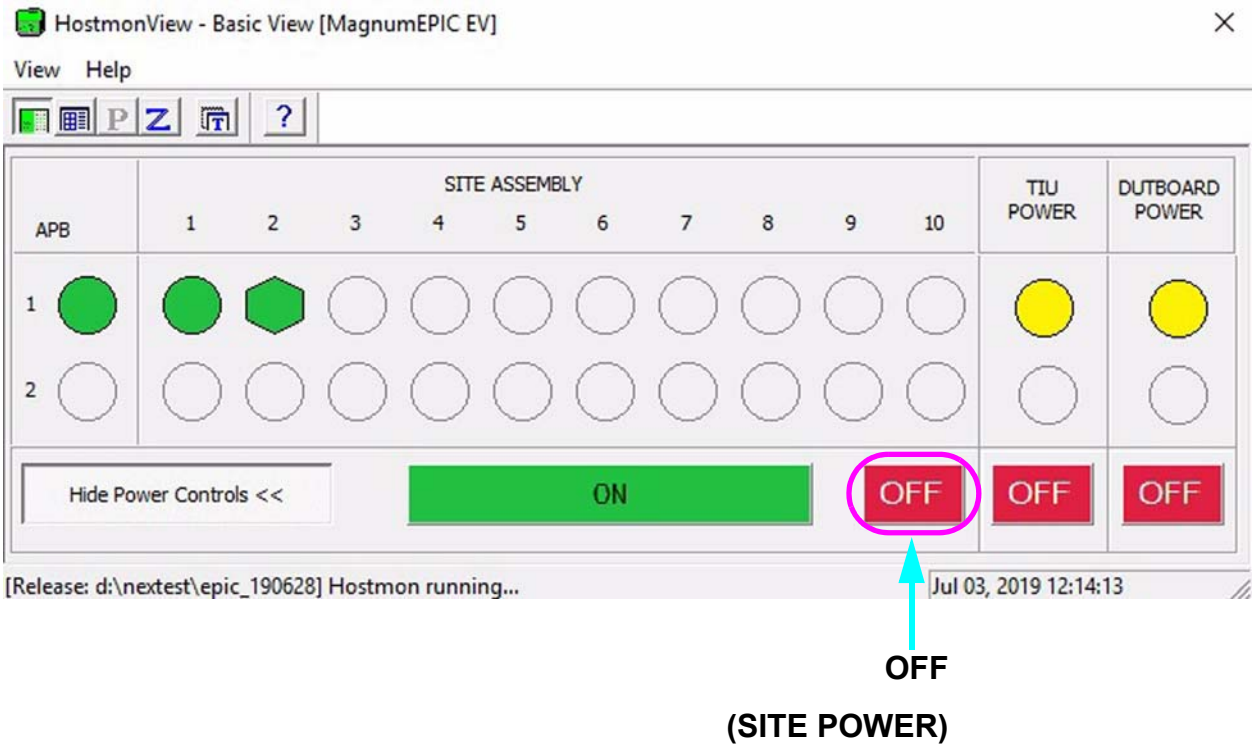
Note: If TIU POWER and/or DUTBOARD POWER are still ON, turning SITE POWER OFF will also turn off TIU POWER and/or DUTBOARD POWER.

Figure-13: SITE Power Off in HostmonView (Magnum EPIC Example)



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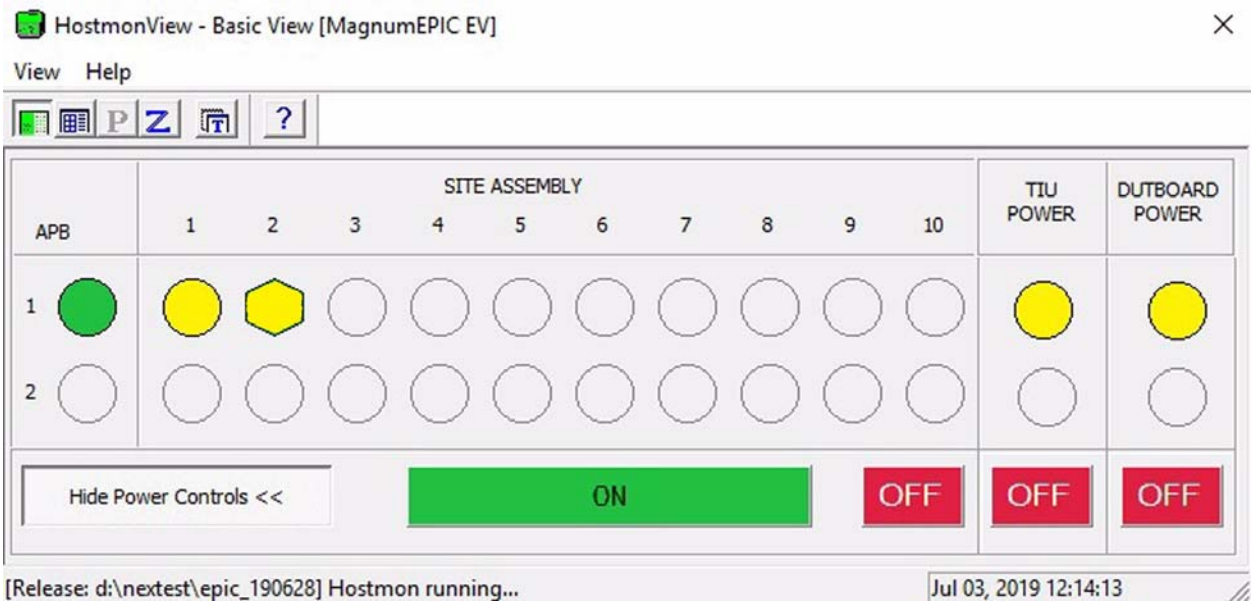
The SITE POWER indicator(s) will display a yellow background in the SITE ASSEMBLY section. This means that SITE power is off. See Figure-14: SITE Power OFF (Magnum EPIC Example) below. See Figure-18: TIU/DUTBOARD Power Indicator Legend, page 33 for other possible DUTBoard Power status.

Figure-14: SITE Power OFF (Magnum EPIC Example)



1.5.3 Power Lock Mode

Note: This section is applicable only to Site Assemblies in Power Lock Mode. Power Lock Mode happens only during the Site boot process.

During the Site boot process, boot programs may have to reconfigure the hardware. When they do this, they put the slot being reconfigured into Power Lock Mode. When a boot module puts a slot into Power Lock Mode, the YELLOW part of the slot icon changes to ORANGE. Power cycling (by turning off system or site power, or opening the Test Head hatch) should NOT be attempted during this critical update time (called Power Lock Mode) since it will render the hardware inoperable.

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Note: Orange = The Site Controller(s) are in Power Lock mode. The Site Controller(s) go into Power Lock mode during the boot sequence.

Therefore, it is recommended that you not click the Site Power OFF button at any time during the site boot process, but especially not when a slot is in Power Lock Mode.

Caution – Powering off site power during Power Lock Mode could severely damage the Site Assemblies and make them nonoperational.

Figure-15: Site 2 in Power Lock Mode (Magnum V-Pre APB/SPC 7.4.1)

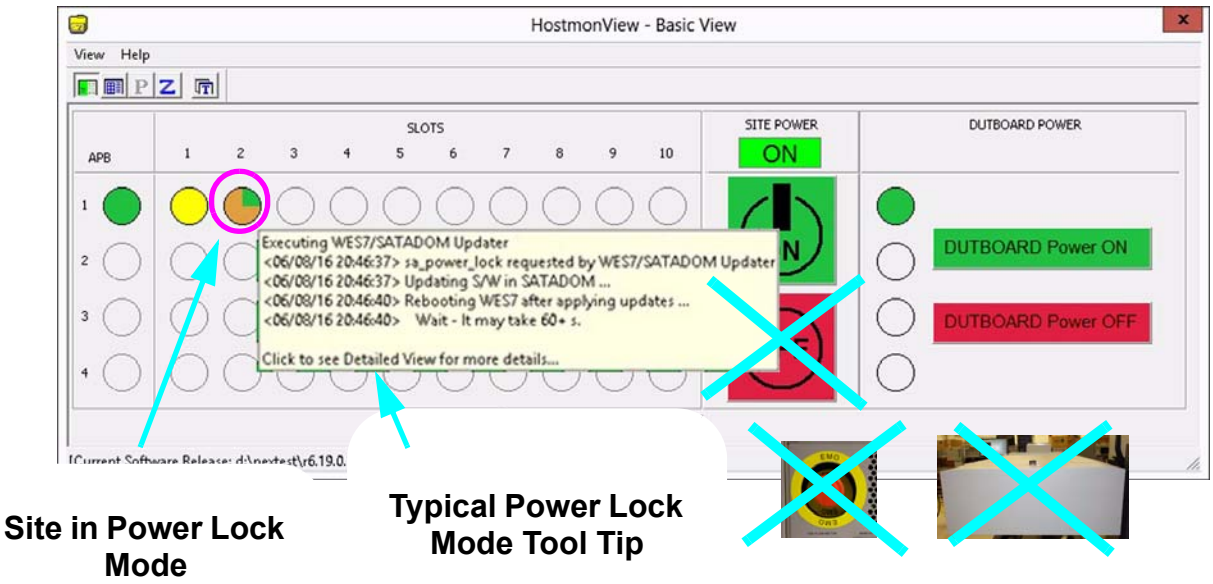

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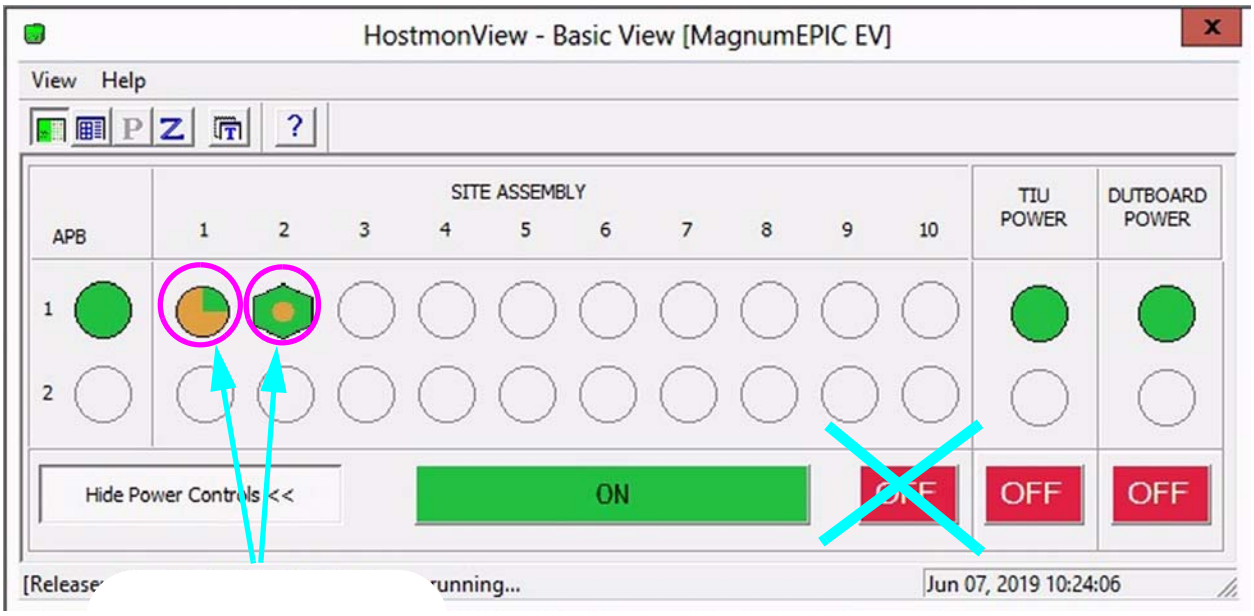
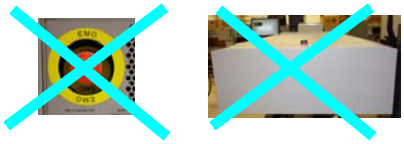
Figure-16: Card Cage 1 Sites in Power Lock Mode (Magnum EPIC)



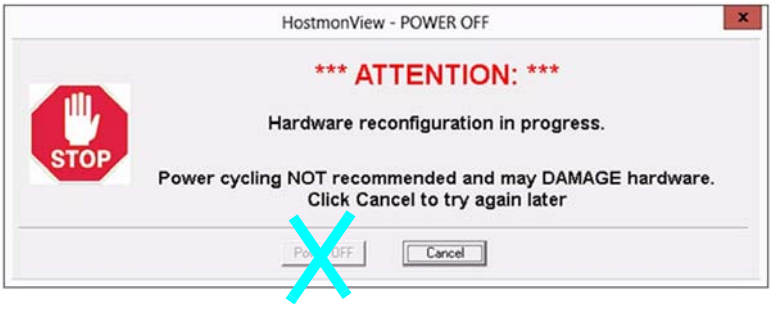
Sites in Power Lock Mode



Moving the cursor over the icon will show the tool tip (displayed in the figure above):

There is an additional safeguard added as a confirmation step. This confirmation appears if the Site Power OFF button is clicked and any of the sites are in Power Lock Mode. This confirmation warns operators of the damage that could be caused by powering off the Sites while they are in Power Lock Mode.

Figure-17: Power Lock Mode Warning Message



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If it is deemed necessary to turn site power off while sites are in Power Lock Mode, the ***ATTENTION*** banner blinks and the **Power OFF** button is enabled 10 seconds after the dialog appears. At this point, site power can be turned off. However, the caution from above will be repeated below for emphasis.

Caution – Powering off site power during Power Lock Mode could severely damage the Site Assemblies and make them nonoperational.

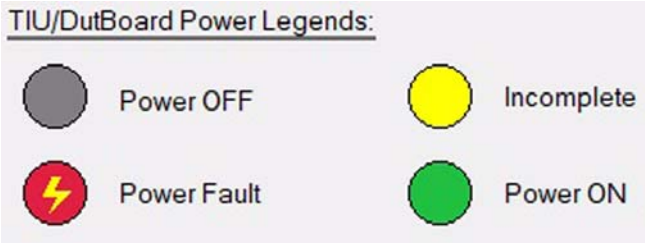
When reconfiguration is completed, the power lock is released, and the ORANGE part of the icon changes back to YELLOW. The Site boot process continues until all of the Site icons are green.

The Site Power Indicator turns green and its label text changes to ON.

1.5.4 DUTBOARD POWER

DUTs often need power while they are being tested, and the Magnum V test systems have a system that provides that power. DUTBoard Power is controlled by the DC Power On/Off Controls (Pre-Firmware 7.4.1) or the DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up) in HostMonView, and there are indicators in HostMonView that indicate whether DUTBoard power is on or off. There is one DUTBoard Power indicator for each APB in the system. Each APB provides power to the DUTs in its section of the tester.

Figure-18: TIU/DUTBOARD Power Indicator Legend



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1.5.5 TIU POWER

Note: If the system being turned OFF is a non-VU Magnum V system, there is no need to perform these instructions. In fact, the button used to perform them will not even appear in HostMonView.

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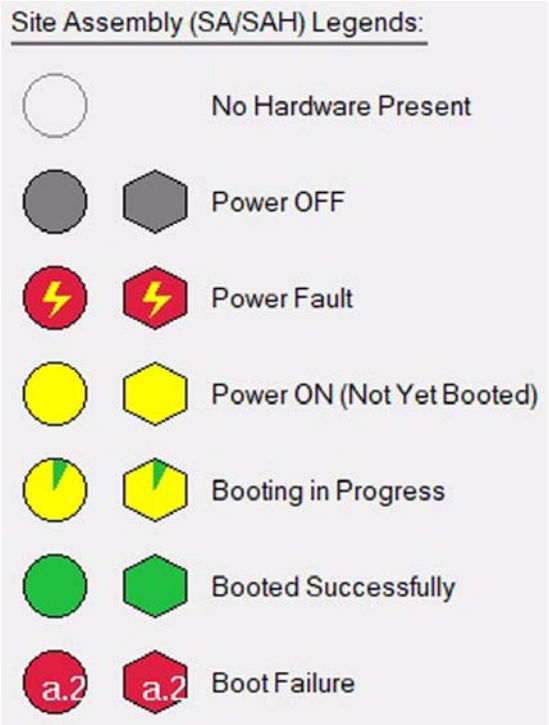
TIUs often need power while DUTs are being tested, and the Magnum VU and EPIC test systems have a system that provides that power. TIU POWER is controlled by the DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up) in HostMonView, and there are indicators in HostMonView that indicate whether TIU POWER is on or off. There is one TIU POWER indicator for each APB in the system. Each APB provides power to the TIUs in its section of the tester.

1.5.6 SITE (ASSEMBLY) Power

SITE Power is controlled by the DC Power On/Off Controls (Pre-Firmware 7.4.1) or the DC Power ON/OFF Controls (Firmware APB/SPC 7.4.1 and Up) in HostMonView, and there are indicators in HostMonView that indicate whether SITE power is on or off. There is one SITE Power indicator for each Site in the system. Each APB provides power to the SITEs in its section of the tester.

Figure-19: Site Assembly Power Indicator Legend



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1.6 Detailed View Display

See Overview, Basic View Display.

The Detailed View display is invoked by:

- Clicking any Status Icon in the Basic View Display.
- Clicking the second of the HostmonView Tool Bar in the Basic View Display:

- Select **View->Detailed**.
- Type `control+D`.

This causes HostmonView to display additional information retrieved from the APB Computers and Site Controllers. This is normally only used by maintenance personnel.

This causes HostmonView to display additional information retrieved from the APB Computers and Site Controllers. The following image shows the key features of this view:

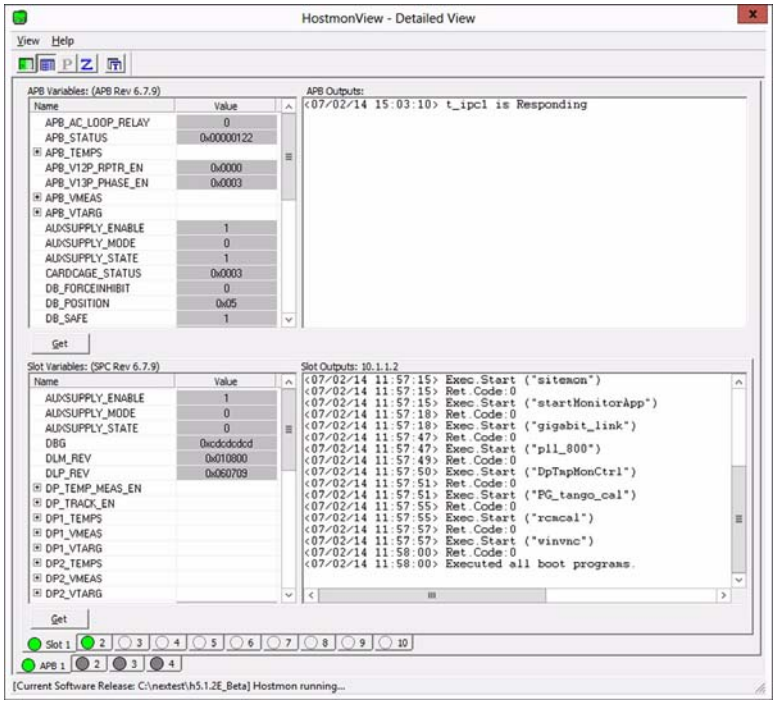

Figure-20: HostmonView Detailed View Display

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Note the following:

- The *Card Cage Select* TABs (bottom left) are used to select one card cage of interest.
- The Variable/Value information in the window above the *Card Cage Select* TABs is automatically retrieved from the APB Computer of the selected chassis. The APB Output window accumulates any messages generated by the APB Computer of the selected chassis.
- A measured value is displayed for each of the Variable/Value parameters which are measured by the APB. Values can be updated by clicking the **Get** button.
- Each Variable/Value parameter includes a status indicator, which come in two types:
  - Fault-only Indicators. These are the first two, labeled *Site Shutdown Request* and *Other APB Fault*. These will be Gray in normal operation but will turn Red if they are the source of a fault.
  - Measured Parameter Indicators. These are all but the first two indicators. These will be Green during normal operation but will turn Red if the APB detects a fault related to the parameter.

:

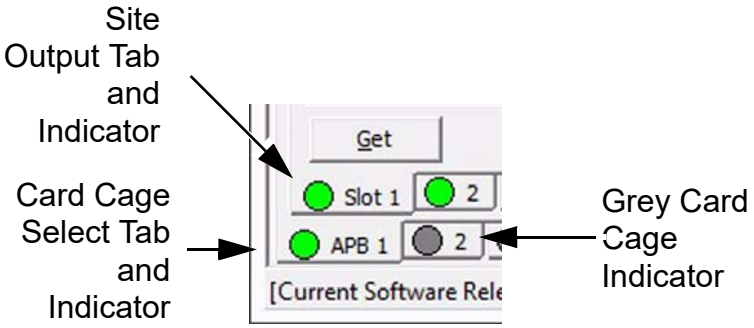

Figure-21: Card Cage Select in Detailed View

- Each *Card Cage Select* TAB has an indicator which will either be Green or Gray:

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Green Card Cage Select Tab and Indicator indicates that the APB Computer of the selected card cage has established TCP/IP communications with the hostmon process and that no fault has been detected.

A Grey Card Cage Indicator either indicates that a chassis' APB Computer is not communicating with the hostmon process. This will normally be because chassis power is off or that the chassis does not exist in the Test System. Note that when power to a given chassis is restored, once that chassis' APB Computer begins to communicate with the hostmon process the corresponding TAB indicator will turn Green (unless a fault is detected).

- The Site Output Tab and Indicators will be Gray, Green, or Red, to indicate the status of the associated Site Assembly in the currently selected card cage.

  Green indicates that the Site Controller has established TCP/IP communications with the hostmon process.

  Gray indicates that the Site Controller is not communicating with the hostmon process, typically because a board is not installed in that slot or because the Site Controller is otherwise not communicating. If there is a fault, Gray indicates that the board is not the source of the fault; i.e. did not send a Shutdown Request to its APB.

- The *Site Output* window displays any status messages generated by the sitemon process executing on the currently selected Site Assembly.
- The *APB Output* window displays any status messages from the APB Computer of the currently selected chassis.

1.7 Prober View Display

See Overview, Basic View Display.

Note: This view is available only in Test Systems that use a Prober.

The Prober View operates and displays information about the Prober System interfacing with the Test System.

Its display is invoked by:

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- Clicking the third button of the HostmonView Tool Bar in the Basic View Display:

This causes HostmonView to display the Prober View.

The following image shows the key features of this view:

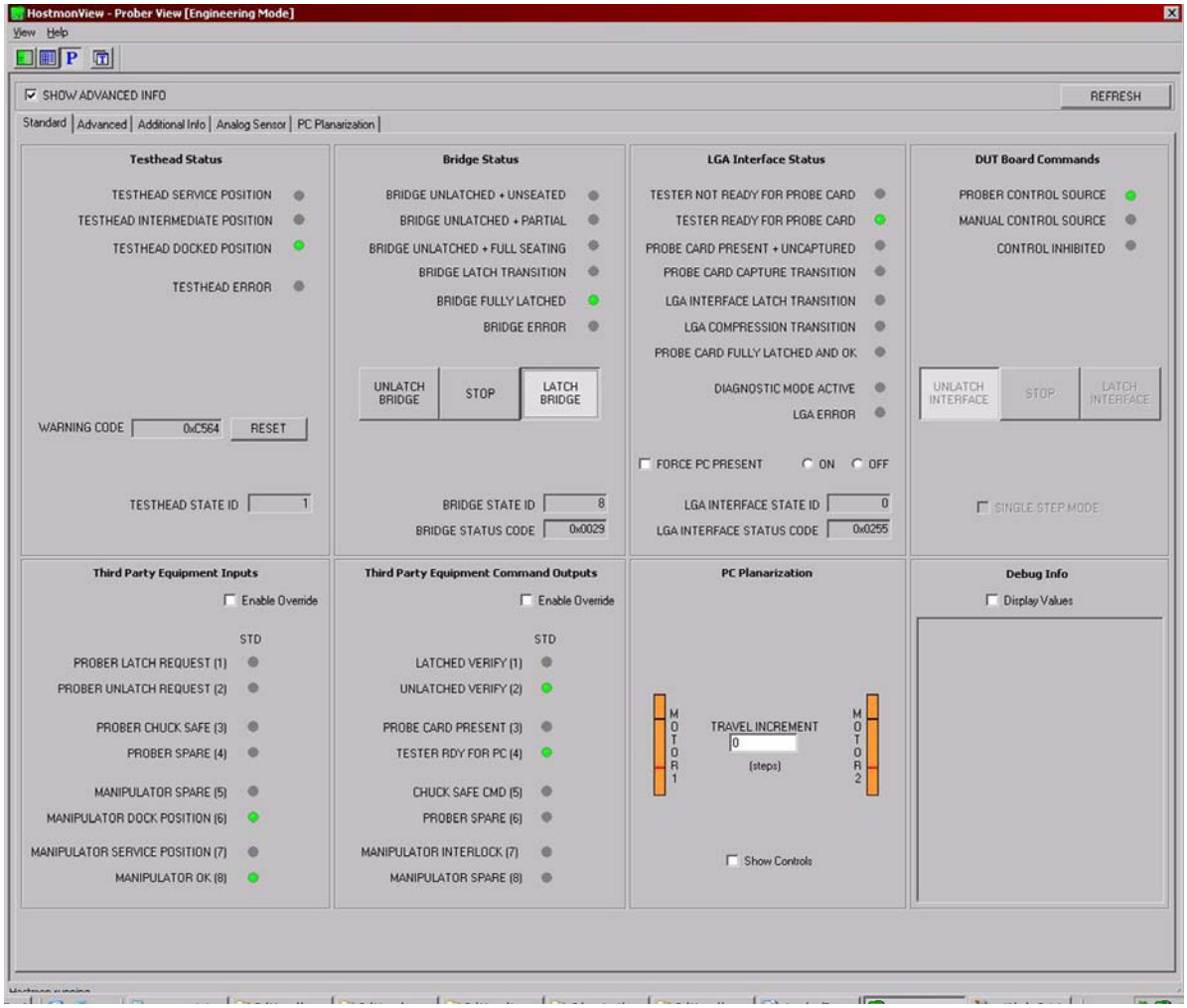

Figure-22: HostmonView Prober View Display

Feature Standard Description:

- **Testhead Status**: Indicates the position of the Test Head.

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- **Bridge Status**: Indicates if the Bridge is latched or unlatched and provides buttons for latching and unlatching the Bridge.
- **LGA Interface Status**: Indicates if the Bridge is latched or unlatched.
- **DUT Board Commands**: Indicates source of control and provides buttons for latching and unlatching the Interface.
- **Third Party Equipment Inputs**: Indicates requests from third party equipment and provides a checkbox to enable the overriding of those commands.
- **Third Party Equipment Command Outputs**: Indicates information from third party equipment and provides a checkbox to enable the overriding of those commands.
- **PC Planarization**: Indicates the status of the manipulator motors.
- **Debug Info**: Displays debug info with a checkbox that enables the display.

1.8 ZIF View Display

See Overview, Basic View Display.

The ZIF View operates and displays information about the Zero Insertion Force (ZIF) connectors. The ZIF connectors connect Site Assemblies with the Tester Interface Unit (TIU).

Its display is invoked by:

- Clicking the fourth of the HostmonView Tool Bar in the Basic View Display:

This causes HostmonView to display Zero Insertion Force (ZIF) connector information retrieved from the APB Computers. This is normally only used by maintenance personnel.

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The following image shows the key features of this view:

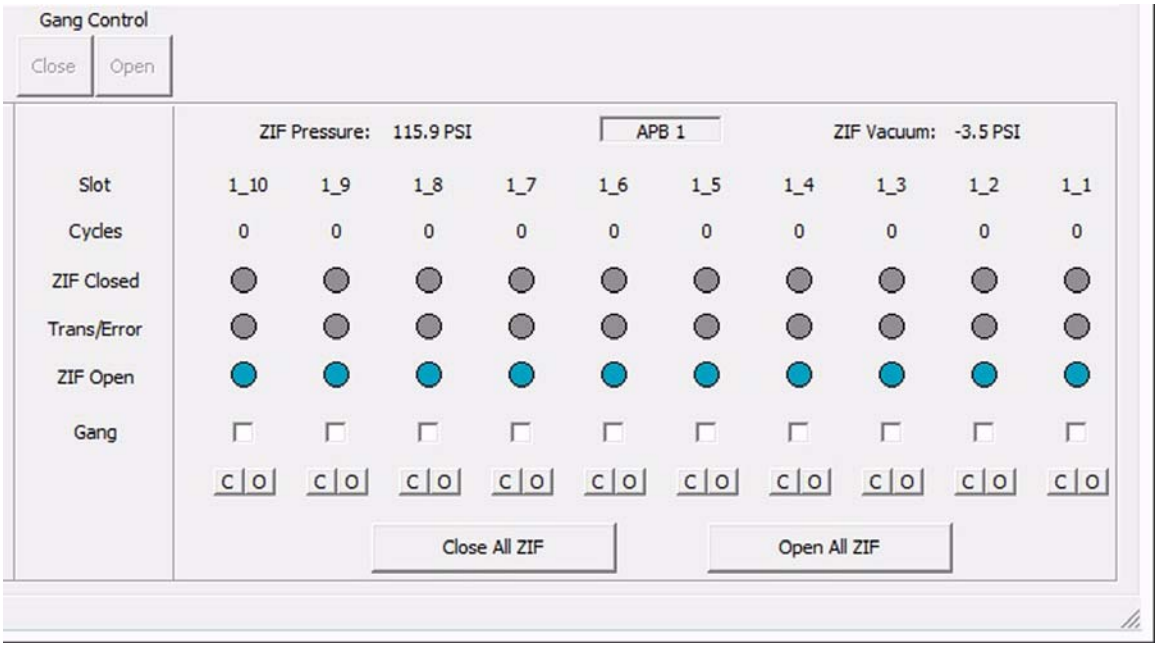

Figure-23: HostmonView ZIF View Display

Feature Description:

- **Gang Control**: Opens or closes the ZIF connectors that have their Gang checkboxes checked.
- **ZIF Pressure**: This is the pressure of the ZIF connectors for this card cage. A pressure of approximately 120 PSI (0.83 MPa) is necessary to hold the ZIF connectors open.
- **APB (Card Cage)**: This is the number of the APB to which this ZIF section pertains. Each card cage is controlled by an APB, so it is also the number of the card cage.
- **ZIF Vacuum**: This is the vacuum of the ZIF connectors for this card cage.
- **Slot**: This is the slot number of the slot for the site assembly that this ZIF connector connects with the TIU. It is also the number of the site assembly. There is one slot and one ZIF connector for every site assembly.
- **Cycles**: Number of ZIF cycles executed.
- **ZIF Closed**: The indicator light for a slot will be illuminate green in this row if its ZIF connector is closed.

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- **Trans/Error**: The indicator light for a slot will be illuminate yellow or red in this row if its ZIF connector is in an indeterminate state due to a lack of pressure (see ZIF Pressure above). Also, the indicator light will illuminate yellow while the ZIF connector is opening or closing, a process that takes a few seconds.
- **ZIF Open**: The indicator light for a slot will be illuminate blue in this row if its ZIF connector is open. NOTE: A pressure of approximately 120 PSI (0.83 MPa) is necessary to hold the ZIF connectors open.
- **Gang**: This control enables all of the checked ZIF connectors to be opened or closed as a group with the Gang Control.
- **C Button**: Clicking the C button for a particular slot closes the ZIF connector for just that slot.
- **O Button**: Clicking the O button for a particular slot opens the ZIF connector for just that slot.
- **Close All ZIF**: Clicking this button closes all the ZIF connectors for the card cage indicated in the APB label.
- **Open All ZIF**: Clicking this button opens all the ZIF connectors for the card cage indicated in the APB label.

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